DATA ANALYSIS · SPRINT 7

Zuber Ride Analysis in Chicago

Exploratory data analysis and statistical hypothesis testing on ride patterns and the impact of weather on trip duration

Python · Pandas · Matplotlib · SciPy · SQL

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CONTEXT

A new ride-sharing operator in Chicago

Zuber is a new ride-sharing company launching in Chicago. Before entering the market, it's essential to understand how the city already moves.

Project goal

Map ride behavior patterns across the city, identify trends among taxi companies and the busiest neighborhoods, and investigate the impact of weather on trip duration.

1

Exploratory analysis

Dataset quality, highest-volume companies, busiest neighborhoods, and graphical visualizations.

2

Hypothesis testing

Relationship between weather and ride duration, comparing rainy and non-rainy Saturdays with Student's t-test.

Data and tools used

`project_sql_result_01.csv`

64 taxi companies

Total ride volume per company

`project_sql_result_04.csv`

94 destination neighborhoods

Average completed rides per neighborhood

`project_sql_result_07.csv`

1,068 rides

Loop → O'Hare, with weather and duration

Data quality

- No missing values in any of the three datasets
- 197 duplicate records removed from the ride dataset before analysis
- start_ts column converted to datetime

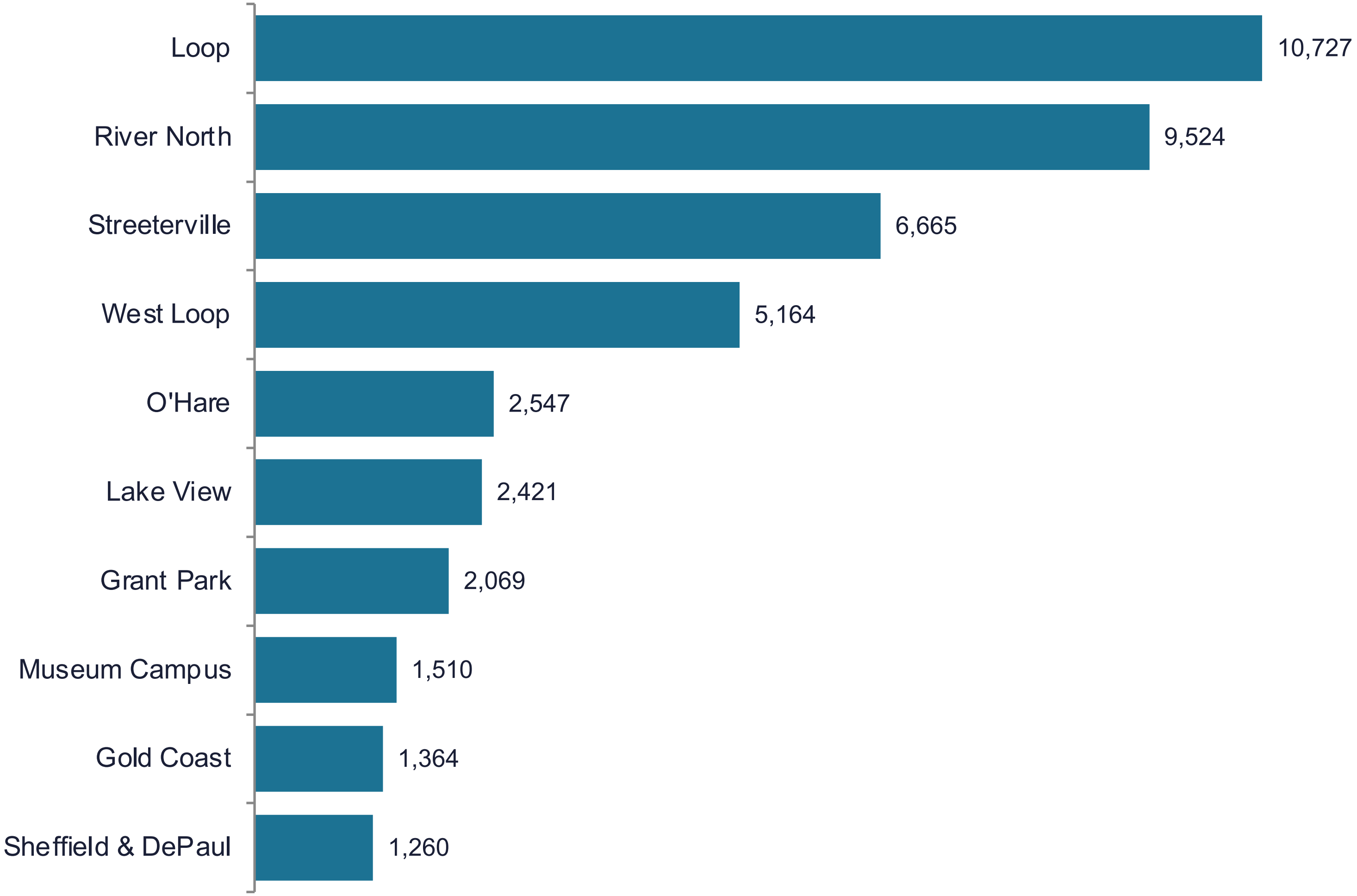
Top 10 taxi companies

19,558

rides by Flash Cab, the clear market leader

The top three companies account for around 30% of total ride volume among the 64 companies analyzed, signaling a market with strong concentration at the top.

Top 10 neighborhoods by ride volume



10,727

average rides to the Loop, the city's financial center

Loop, River North, Streeterville, and West Loop concentrate the highest demand, reflecting central areas with strong commercial and tourist activity.

HYPOTHESIS TESTING

Does weather influence ride duration?

Scope: rides between the Loop and O'Hare International Airport on Saturdays, comparing bad weather (rain) and good weather days.

H_0 · Null hypothesis

The average ride duration on rainy Saturdays equals the average ride duration on non-rainy Saturdays.

H_1 · Alternative hypothesis

The average ride duration on rainy Saturdays differs from the average ride duration on non-rainy Saturdays.

Student's t-test

independent samples, unequal variances

$\alpha = 0.05$

adopted significance level

148 vs. 723

rides in bad weather vs. good weather

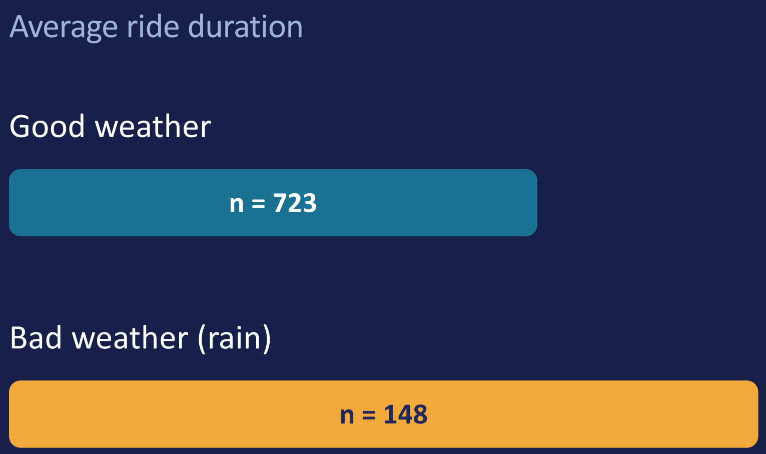
TEST RESULT

H_0 rejected: weather changes ride duration

$$p = 9.13 \times 10^{-8}$$

far below $\alpha = 0.05$

The statistical evidence indicates that the average ride duration between the Loop and O'Hare Airport on Saturdays differs on bad weather days compared to favorable weather days.



Sample sizes differ, so the t-test was applied without assuming equal variances between groups.

CONCLUSION

What this means for Zuber

1

Concentrated market

A handful of companies, like Flash Cab, dominate ride volume. There is room for differentiation against smaller operators.

2

Regional demand

Loop, River North, Streeterville, and West Loop concentrate the highest demand and should guide fleet prioritization.

3

Weather as a variable

Weather affects ride duration on airport routes, suggesting pricing and ETA adjustments should be made on rainy days.

Suggested next step: cross-reference peak hours by neighborhood with weather history to anticipate demand spikes and travel time.